

Developer Setup Guide

Git • GitHub • VS Code • Extensions

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01 Git — Initial Setup

What is Git?

Git is a free, open-source distributed version control system that tracks changes in source code. It allows multiple developers to collaborate efficiently on the same project.

Step 1: Download & Install Git

1. **Windows:** Visit <https://git-scm.com/download/win> → Download latest installer → Run .exe → follow wizard (keep all defaults).
2. **macOS:** Open Terminal: `xcode-select --install` OR download from <https://git-scm.com/download/mac>
3. **Linux (Ubuntu/Debian):** Open Terminal and run the commands below.

```
# Linux Installation
sudo apt update
sudo apt install git

# Verify (all OS)
git --version
```

Step 2: Configure Git Identity

Set your name and email — these are attached to every commit you make.

```
git config --global user.name "Your Full Name"
git config --global user.email "you@example.com"
# Verify:
git config --list
```

Step 3: Set Default Branch & Editor

```
# Set 'main' as default branch
git config --global init.defaultBranch main
# Set VS Code as default editor
git config --global core.editor "code --wait"
# Line endings - Windows:
git config --global core.autocrlf true
# Line endings - macOS/Linux:
git config --global core.autocrlf input
```

Step 4: Essential Git Commands — Quick Reference

Command	Description
<code>git init</code>	Initialize a new local repository
<code>git clone <url></code>	Copy a remote repo to your machine
<code>git status</code>	Show changes in working directory
<code>git add .</code>	Stage all changed files
<code>git commit -m "msg"</code>	Save staged changes with a message

git push	Upload commits to remote repo
git pull	Download and merge latest changes
git branch	List all branches
git checkout -b <n>	Create and switch to a new branch
git merge <branch>	Merge a branch into current branch
git log --oneline	View compact commit history

What is GitHub?

GitHub is a cloud-based hosting platform for Git repositories used by over 100 million developers. It provides collaboration tools, code review, issue tracking, and CI/CD pipelines.

Step-by-Step: Creating a GitHub Account

4. **Visit GitHub:** Open <https://github.com> in your browser.
5. **Click Sign Up:** Click the green "Sign up" button in the top-right corner.
6. **Enter Email:** Use the same email you configured in Git above.
7. **Create Password:** Strong password — mix of uppercase, lowercase, numbers, symbols.
8. **Choose Username:** Professional and unique. It appears in your URL:
`github.com/yourusername`
9. **Solve CAPTCHA:** Complete the verification puzzle.
10. **Verify Email:** Enter the 6-digit code sent to your inbox.
11. **Complete Setup:** Answer onboarding questions or skip — your account is now ready.

SSH Key Setup (No Password Needed)

```
ssh-keygen -t ed25519 -C "you@example.com"
# Press Enter to accept defaults
# Copy public key:
# Windows: cat ~/.ssh/id_ed25519.pub | clip
# macOS: pbcopy < ~/.ssh/id_ed25519.pub
# Linux: cat ~/.ssh/id_ed25519.pub
```

12. **On GitHub:** Settings → SSH and GPG keys → New SSH key → paste key → Add SSH key.
13. **Test:** `ssh -T git@github.com` should respond: Hi username!

Creating Your First Repository

14. **Click '+':** Top-right of GitHub → New repository.
15. **Name:** Use hyphens, no spaces (e.g., my-first-project).
16. **Visibility:** Public or Private.
17. **Initialize:** Check 'Add a README file' → Create Repository.

```
git clone git@github.com:yourusername/your-repo-name.git
cd your-repo-name
```

03 VS Code — Installation

What is VS Code?

Visual Studio Code (VS Code) is a free, lightweight, and powerful source code editor by Microsoft. It supports hundreds of languages and has a rich extension ecosystem.

Download & Install

18. **Official Site:** Visit <https://code.visualstudio.com> (always use the official site only).
19. **Windows:** Run the .exe → Accept license → check 'Add to PATH' → Install.
20. **macOS:** Open .dmg → Drag VS Code to Applications folder → launch from there.
21. **Linux:** `sudo snap install code --classic`

Important Settings After Install

22. **Sign In:** File → Preferences → Accounts → Sign in with GitHub (syncs settings).
23. **Auto Save:** File → Auto Save.
24. **Word Wrap:** View → Word Wrap (or Alt+Z).
25. **Terminal:** View → Terminal (or Ctrl+`) — built-in terminal.

Key Shortcuts

Shortcut	Action
Ctrl + P	Quick Open — search and open any file
Ctrl + Shift + P	Command Palette — run any VS Code command
Ctrl + `	Toggle integrated terminal
Ctrl + /	Comment / Uncomment selected lines
Alt + Shift + F	Format entire document
Ctrl + D	Select next occurrence of word
Ctrl + B	Toggle sidebar visibility
F5	Run / Debug current file
Ctrl + Shift + G	Open Source Control (Git) panel

04 VS Code Extensions — Mandatory & Optional

How to Install Extensions

26. Click the Extensions icon in the left sidebar (or Ctrl+Shift+X).
27. Search the extension name, verify the publisher, click Install.
28. Click Reload if prompted after installation.

AI-Powered Extensions (Copilot)

GitHub Copilot

AI pair programmer — suggests code completions, full functions, and docs as you type. FREE for students via GitHub Education Pack. Publisher: GitHub | ID: GitHub.copilot

How to Enable Copilot:

29. Install GitHub Copilot extension.
30. Click Copilot icon in bottom status bar → Sign in to GitHub.
31. Authorize VS Code in the browser pop-up.
32. Copilot starts suggesting code automatically. Press Tab to accept, Esc to dismiss.

Copilot Chat

GitHub Copilot Chat — AI chat inside VS Code. Ask questions about your code, get debugging help, explanations, and code generation. Publisher: GitHub | ID: GitHub.copilot-chat

Mandatory Extensions for Every Developer

Extension	Purpose & Publisher
Prettier	Auto-formats code on save. Publisher: Prettier (esbenp.prettier-vscode)
ESLint	Catches JS/TS errors and coding standard violations. Publisher: Microsoft
GitLens	Inline Git blame, history, visual diff. Publisher: GitKraken
Git Graph	Visual graph of your Git repository. Publisher: mhutchie
Path IntelliSense	Auto-completes file paths in code.
Auto Rename Tag	Auto-renames paired HTML/JSEX closing tags.
Error Lens	Shows errors and warnings inline — no hover needed.
Live Server	Local development server with live reload.
indent-rainbow	Colors indentation levels for easy reading.

Language-Specific Extensions

Language	Recommended Extension
Python	Python (Microsoft) — IntelliSense, debugging, Jupyter
Java	Extension Pack for Java (Microsoft)
HTML / CSS	HTML CSS Support + IntelliSense for CSS class names
React / JSX	ES7+ React/Redux Snippets (dsznajder)
SQL / Database	SQLTools + your DB driver
Markdown	Markdown All in One (Yu Zhang)
REST API Testing	Thunder Client — Postman inside VS Code
Docker	Docker (Microsoft)

Prettier Configuration (settings.json)

33. Press Ctrl+Shift+P → type "Open Settings (JSON)" → Enter.

34. Add the following configuration:

```
{
  "editor.defaultFormatter": "esbenp.prettier-vscode",
  "editor.formatOnSave": true,
  "editor.tabSize": 2,
  "files.autoSave": "onFocusChange",
  "editor.wordWrap": "on",
  "github.copilot.enable": { "*": true }
}
```




Final Setup Checklist

Complete Setup Verification

Task	Verification
Git installed	git --version shows a version number
Git user name configured	git config user.name shows your name
Git email configured	git config user.email shows your email
GitHub account active	Login at github.com works successfully
SSH key added to GitHub	ssh -T git@github.com says: Hi username!
VS Code installed	code --version in terminal shows version
GitHub Copilot active	Copilot icon visible in VS Code status bar
Prettier installed	Alt+Shift+F formats the file correctly
GitLens installed	Inline blame text appears next to code lines
Live Server installed	Right-click HTML → Open with Live Server works

TIP

Students: Apply for the GitHub Student Developer Pack at education.github.com/pack to get GitHub Copilot FREE along with many premium developer tools.

NOTE

Always download Git and VS Code from their official websites only. Avoid third-party sites to prevent malware.